Sahan S Liyanage

Solar and Space Physics Researcher | Scientific Software Developer

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Physics researcher working across solar radio astronomy, coronal shock waves, space weather, observational solar physics, and scientific software. Current work connects automated observations, physical models, and reproducible computation.

EDUCATION

Bachelor of Science Honours in Physics - Second Class (Upper Division)

Jan 2020 - May 2024

University of Colombo, Sri Lanka

- Cumulative GPA: 3.57/4.00 (126 credits; 252 ECTS). Physics GPA: 3.65/4.00 (66 credits; 132 ECTS).
- Relevant coursework included quantum mechanics, classical mechanics, electromagnetic fields, statistical physics, advanced optics, solid-state physics, nuclear and particle physics, computational physics, electronics, data acquisition, signal processing, and advanced physics laboratories.

PROFESSIONAL EXPERIENCE

Research Assistant (Contract)

May 2026 - Present

Arthur C. Clarke Institute for Modern Technologies, Sri Lanka

- Project: Spectropolarimetric observations of band-split Type II solar radio bursts using a long-wave antenna.
- Contribute to an integrated CALLISTO and H-alpha solar-observation system.
- Work on automated acquisition, archiving, monitoring, and visualization of solar-radio and optical observations.
- Support a real-time monitoring dashboard for CALLISTO dynamic spectra and H-alpha imaging data.

RESEARCH EXPERIENCE

Research Assistant

Mar 2025 - Present

Astronomy and Space Science Unit, University of Colombo

Analysis of Metric Type II Solar Radio Burst Drift Rates and Their Space-Weather Connections Using a GUI Tool for e-CALLISTO

- Study metric Type II solar-radio-burst drift rates and links with CMEs, solar flares, solar energetic-particle events, and geomagnetic storms.
- Analyze e-CALLISTO dynamic spectra and develop a Python GUI for FITS reading, processing, merging, visualization, and physical analysis.
- Supervisor: Dr. Janaka Adassuriya.

Undergraduate Research Project

May 2023 - Apr 2024

Department of Physics, University of Colombo

Determination of the Speed and Source Height of Coronal Shock Waves Using Type II Solar Radio Bursts

- Analyzed 25 metric Type II solar-radio-burst datasets from the e-CALLISTO network.
- Developed a Python workflow for FITS reading, background-noise reduction, burst isolation, and dynamic-spectrum analysis.
- Derived shock-speed and source-height expressions using plasma-oscillation theory and the Newkirk coronal-density model.
- Estimated shock speeds of 343-1032 km/s and source heights of 1.317-1.724 solar radii.
- Supervisors: Dr. Janaka Adassuriya and Prof. K. P. S. C. Jayaratne.

Raman Spectroscopy Project

Nov 2022 - Mar 2023

Department of Physics, University of Colombo

Raman Spectroscopic Analysis of Common Minerals in Sri Lanka

- Analyzed calcite, dolomite, apatite, quartz, feldspar, graphite, and mica using a 532 nm Raman system.
- Prepared samples and processed spectra in Python using calibration, baseline correction, smoothing, and plotting.
- Supervisor: Prof. H. H. E. Jayaweera.

Undergraduate Research Mentor

May 2025 - Apr 2026

Final-year project mentoring

Estimating Coronal Magnetic Field Strength from Band Splitting in Type II Solar Radio Bursts

SCIENTIFIC SOFTWARE

e-CALLISTO FITS Analyzer

Active

Open-source, cross-platform scientific application

- Python and PySide6 application for Windows, macOS, and Linux.
- Supports archive acquisition, time/frequency merging, background subtraction, polygon burst isolation, drift-rate fitting, shock-parameter estimation, GOES X-ray visualization, SOHO/LASCO CME browsing, and scientific export.
- github.com/SaanDev/e-Callisto_FITS_Analyzer

PUBLICATIONS

Liyanage, G. L. S. S., Adassuriya, J., Jayaratne, K. P. S. C., & Monstein, C. (2025). Determination of speed and source height of coronal shock waves using Type II solar radio bursts. *Proceedings of the Technical Sessions*, 41, 59-67. Institute of Physics, Sri Lanka.

Liyanage, G. L. S. S., Adassuriya, J., Jayaratne, K. P. S. C., Monstein, C., & Manoharan, P. K. (2026). e-CALLISTO FITS Analyzer: A Software Framework for CALLISTO Solar Radio Data. Manuscript submitted for publication.

SCIENTIFIC PRESENTATIONS

Determination of Speed and Source Height of Coronal Shock Waves Using Type II Solar Radio Bursts

22 Mar 2025

41st Technical Sessions, Institute of Physics, Sri Lanka - University of Colombo

- On-site oral presentation.

Solar Radio Bursts and Space Weather

20 Nov 2024

Physics Society, University of Colombo

- On-site invited presentation to physics students.

TEACHING EXPERIENCE

Teaching Assistant

Aug 2025 - Mar 2026

University of Colombo School of Computing

- Probability and Statistics for first-year computer-science majors.
- Digital Signal Processing for second-year students.
- Cryptography and Information Security for second-year students.

Teaching Assistant

May 2024 - May 2025

Department of Physics, University of Colombo

- Computational Electrodynamics and Advanced Physics Laboratory I.
- Analogue and Digital Electronic Principles Laboratory.
- General Physics Laboratory I and II.

WORKSHOP AND TRAINING

Participant - COSPAR Capacity Building Workshop 2024
19-30 Aug 2024

Samarkand State University, Samarkand City, Uzbekistan

- Training on coronal and interplanetary shocks using SOHO, SDO, STEREO, and ground-based observations with SolarSoft, SunPy, AstroPy, and Python.

TECHNICAL SKILLS

Programming and computation: Python, MATLAB, C/C++, machine learning, OriginPro, LaTeX, web development, Linux, Microsoft Office.

Scientific and technical: FITS data, solar-radio dynamic spectra, Raman spectral processing, Arduino, Adobe Photoshop, data acquisition, signal processing, scientific visualization, and interactive desktop software.

LEADERSHIP

Vice President
2022-2023

Astronomical Society, University of Colombo

Coordinator
2020-2022

Astronomical Society, University of Colombo

REFERENCES

Professional references are available on request. Referee contact details are intentionally omitted from this public web edition.